Predicting hourly bike-sharing demand with a chained regression–classification pipeline

Machine Learning, Winter 2026 — Project 1. Team seed 36954.

1. Problem and data

The task is to predict the number of bikes rented per hour (*cnt*) and, finally, to classify hours as high demand. The labelled file contains 10,886 hours over 456 days (days 1–19 of each month, 2011–2012); the hidden test file contains 574 hours, all on day 20. The team seed, the SHA-256 of the sorted, underscore-joined student IDs modulo 100,000 (the brief's example reproduces 41,698), splits the labelled rows into 8,708 training and 2,178 validation hours. Every encoder, scaler and centring mean is fitted on the training portion only. Three properties shape the design: *workingday* is an exact function of *weekday* and *holiday*, *season* is an exact function of *mnth* in this file, and the year, month and weekday of *dteday* coincide exactly with *yr*, *mnth* and *weekday*. Phases are chained through saved artifacts. Phases 2–5 were preceded by written expectations; those of Phase 1 and the bonus were reconstructed afterwards (Section 8). Besides R^2 and RMSE, every regression model is checked on the hours that matter to the operator: the mean shortfall in the top demand decile (actual demand above 452 bikes), grouped both by actual and by predicted demand.

2. Phase 1 — Gradient descent

Method. A linear model with main effects only was fitted by full-batch gradient descent on $L(w) = \|Xw - y\|^2/(2n)$, with gradient $X^T(Xw - y)/n$. Hour, weekday, weathersit (level 4, absent from the training portion, merged into 3) and month were one-hot encoded with one reference level each; *yr*, *holiday*, *temp*, *hum* and *windspeed* enter as numbers, and *workingday*, *season*, *atemp* and *instant* were excluded as exact or near copies. Month was preferred to season (validation R^2 0.6823 vs 0.6805); the elapsed-day trend from *dteday* added only 0.0005 R^2 while raising the condition number from 56 to 8,084 (36,407 instead of 512 updates) and was left out. Because the loss is quadratic, gradient descent is stable only for learning rates below $2/\lambda_{max}$, where $\lambda_{max} = 1.9894$ is the largest eigenvalue of X^TX/n. Learning rates of 0.005–$2.1/\lambda_{max}$ were compared (Fig. 1); convergence was declared when $\max|\text{gradient}| < 10^{-4}$ (cap 50,000 updates). The loop uses the equivalent precomputed form $Hw - b$ with $H = X^TX/n$ and $b = X^Ty/n$, and asserts at the start and end of every fit that both forms give the same gradient.

Results. $2.1/\lambda_{max}$ diverged after 84 updates; $0.005/\lambda_{max}$ reached the cap with max|gradient| 0.0104; all rates between converged to the same loss. The chosen rate $1.9/\lambda_{max} = 0.955$ converged in 267 updates, within 0.0074 bikes of the exact least-squares solution. Training R^2 0.6970, validation R^2 0.6823 (RMSE 103.6): the model underfits. Hours above 452 bikes were under-predicted by 176.6 bikes on average, and even the hours predicted to be busiest by 95.9; 213 validation hours were predicted below zero. **Passed on:** the weight vector, learning rate, update count, feature list and fitted scaler; 44.4% of the training residual variance lay between hour × working-day cells.

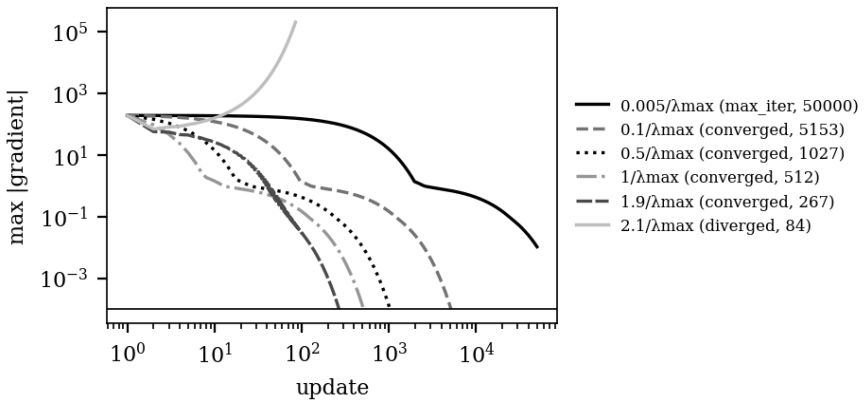


Figure 1. Phase 1 learning-rate sweep: max|gradient| against update (log–log); horizontal line: tolerance.

3. Phase 2 — Polynomial regression

Method. Interaction groups were chosen from Phase 1's training residuals, ranked by the share of residual variance lying between groups and re-ranked after removing the confirmed groups. Degree is interpreted as interaction order: degree 2 adds hour × workingday, hour × yr, hour × season, hour-block × weathersit and temp × hum; degree 3 adds hour × workingday × yr and hour × workingday × temp with their

lower-order terms. Terms are explicit lists (no products of two levels of one column, no squares of binary columns); every non-hour factor is centred at its training mean before multiplying, which lowered the condition number of the full design from 21,616 to 1,713 without changing what it can express. Each group was added in turn, every fit warm-started from the Phase 1 weights (new weights zero, predictions asserted identical) and run with the same $1.9/\lambda_{max}$ rule.

Results. hour × workingday alone raised validation R^2 from 0.6823 to 0.8310; degree 2 reached 0.9009 and degree 3 0.9225 (RMSE 51.1, 242 features, 3,884 updates). The train–validation R^2 gap was -0.0047, with near-equal RMSE, i.e. sampling noise. The top-predicted-decile shortfall fell from 134.9 to 43.3 bikes. temp × hum added only 0.0005 R^2. **Passed on:** degree 3, the 242-feature list with lineage, weights and scaler.

4. Phase 3 — Bias–variance trade-off

Method. The sweep followed Phase 2's order and continued beyond it (hour × holiday, hour × weekday, hour × month, full hour × weathersit and a degree-4 group hour × workingday × yr × season) to reach the high-variance side; columns that were exact combinations of earlier ones were detected by QR factorisation and dropped. Learning curves were drawn for the degree-3 and the largest design. The one chronological split held out the last 20% of labelled days (from 2012-08-05, whole days), refitting all preprocessing on the earlier days. The target rule, fixed beforehand, took the converged fit with the fewest features within 0.002 of the best seeded validation R^2.

Results. Seeded validation R^2 rose from 0.9225 to a plateau of 0.9274 at 818 features while the gap widened from -0.0047 to 0.0112 (Fig. 2). The degree-3 learning curve was flat with train ≈ validation RMSE (bias-limited); the largest design showed a gap of 26.0 bikes at 25% of the data, closing to 4.8 (variance-limited). Chronologically, degree 3 scored R^2 0.9094 (RMSE 65.5); the extensions up to full hour × weathersit improved it to 0.9243, but degree 4 fell to 0.9078, because no 2012-autumn hour exists in the earlier days. **Diagnosis:** degree 3 is bias-limited; variance appears only at degree 4. **Passed on:** the 564-feature design up to hour × month (seeded R^2 0.9262, chronological 0.9224) and the noise candidate temp × hum.

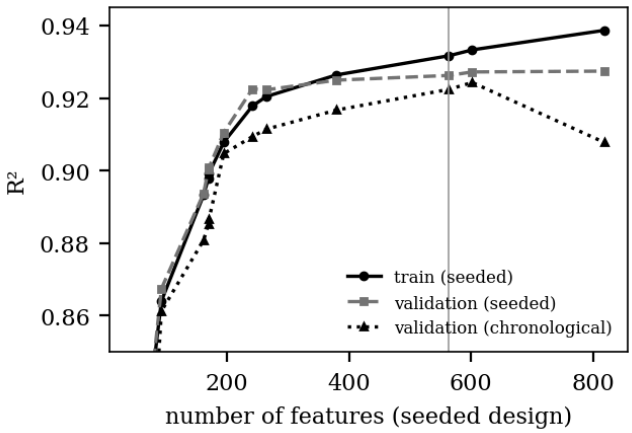


Figure 2. Phase 3 complexity sweep; vertical line: the 564-feature target.

5. Phase 4 — Regularisation

Method. One objective was used for all methods, $J(w) = \|Xw - y\|^2/(2n) + \lambda[\rho\Sigma|w_j| + (1-\rho)\Sigma w_j^2/2]$ with the intercept unpenalised; Ridge ($\rho = 0$), Lasso ($\rho = 1$) and Elastic Net ($\rho \in \{0.1, 0.5, 0.9\}$). Each update takes a gradient step on the smooth part (the Ridge term adds $\lambda(1-\rho)$ to the diagonal of H) and then soft-thresholds: every penalised weight is moved towards zero by $\eta\lambda\rho$ and set to exactly zero if it would cross it. The step is $1.9/L$ for Ridge and $1/L$ with an L1 part ($L = \lambda_{max} + \lambda(1-\rho)$), so the objective never increases. Each method was fitted along a log-spaced λ path from its largest value downwards, every fit warm-started from the previous one; selection used the seeded validation set alone (no k-fold, as the brief fixes one holdout): Ridge by best R^2, Lasso and Elastic Net by the largest λ within 0.002 of the path's best.

Results. All L1-type paths first peaked at the last grid value, so they were extended to $\lambda = 10^{-5}$ at the same spacing; the first 30 values reproduced exactly and only the Elastic Net choice changed. Ridge was best ($\lambda = 0.00045$, R^2 0.926237, equal to the unpenalised fit); Lasso ($\lambda = 0.169$) kept 466 of 564 weights at R^2 0.9248; Elastic Net chose $\rho = 0.1$ ($\lambda = 0.0144$, R^2 0.9246, 564 nonzero). With no variance left to remove, the penalty only added bias, and it raised the top-decile shortfall from 53.8 to 56.9 (Lasso) bikes (Fig. 3). Column verdicts (Table 1) combine a grouped ablation — the design rebuilt without the column and every term derived from it, refitted with Ridge — or an add-back test, the R^2 of the column alone in its full

encoding, and Lasso survival: *useful* if removing it costs at least 0.002 R^2; otherwise *redundant* if alone $R^2 \geq 0.01$; otherwise *uninformative*. Holiday keeps most of its terms under Lasso yet costs nothing when removed: a surviving coefficient is not evidence of usefulness. **Passed on:** the 466 Lasso survivors (from 11 original columns) with lineage, and Ridge as the recommended regression model.

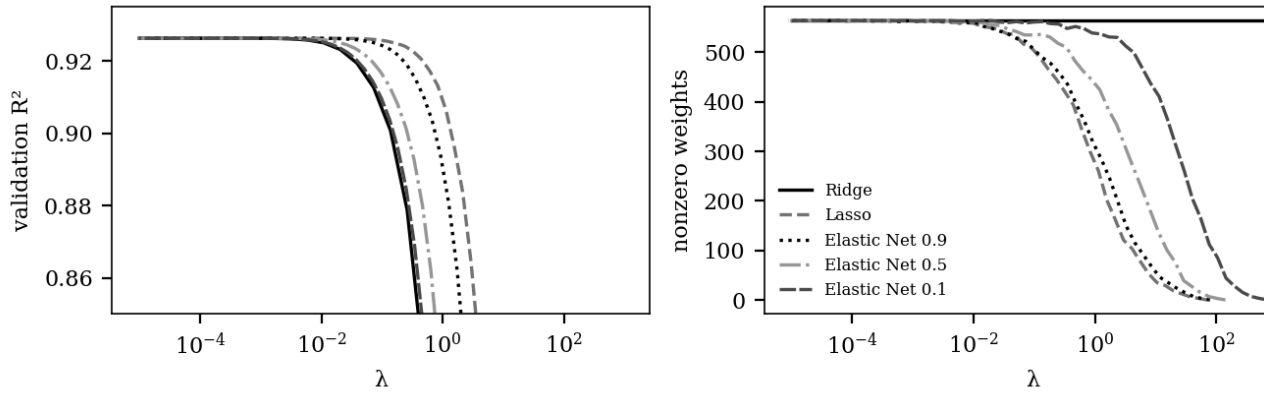


Figure 3. Phase 4 regularisation paths: validation R^2 and number of nonzero weights against λ.

column	test	ΔR^2	alone R^2	Lasso kept	verdict
instant	add back	0.0003	0.094	–	redundant
dteday	add back	0.0002	0.094	–	redundant
season	ablation	0.0000	0.053	56/69	redundant
yr	ablation	−0.1103	0.067	45/48	useful
mnth	ablation	−0.0055	0.070	149/195	useful
hr	ablation	−0.5684	0.491	443/537	useful
holiday	ablation	−0.0000	−0.000	18/24	uninformative
weekday	ablation	−0.0052	−0.001	100/121	useful
workingday	ablation	−0.0162	0.000	63/71	useful
weathersit	ablation	−0.0175	0.016	9/10	useful
temp	ablation	−0.0143	0.165	42/49	useful
atemp	add back	0.0003	0.162	–	redundant
hum	ablation	−0.0006	0.097	2/2	redundant
windspeed	ablation	−0.0006	0.015	1/1	redundant

Table 1. Column verdicts (validation; ΔR^2 relative to the Ridge model).

6. Phase 5 — Logistic regression

Method. An hour is labelled high demand if its count exceeds the 75th percentile of training hours in the same hour × working-day × year group (96 cutoffs from the training portion, applied unchanged to validation). A global cutoff would label eight hours of the day as never high, and a cutoff without year would label 2011 and 2012 hours very differently. 2,127 training hours (24.4%) are positive (25.2% in validation). The inputs are exactly the 466 Lasso survivors with the fitted scaler. The log-loss was minimised by gradient descent with gradient $X^T(p - y)/m$ computed explicitly at each step; since its curvature is at most $\lambda_{max}(X^TX/m)/4 = 1.0424$, the step $1.9/L = 1.823$ is safe. The probability cutoff maximises validation F1 over 0.05–0.95.

Results. The fit converged in 1,229 updates. ROC-AUC 0.871; at the tuned cutoff 0.37: accuracy 0.820 (always-0 baseline 0.748), precision 0.628, recall 0.707, F1 0.665; confusion matrix TN 1399, FP 230, FN 161, TP 388. At 0.5 accuracy is higher (0.827) but F1 lower (0.621, recall 0.563): with a quarter of the hours positive, accuracy rewards missing busy hours (Fig. 4). Temperature has the largest odds ratio (×3.65 per standard deviation), followed by month dummies. The label still carries seasonality (0.4% of January but 48.8% of June training hours are positive): a calendar-only ranking reaches ROC-AUC 0.809, and within a month the classifier reaches 0.747 (within a season 0.641–0.862), which measures its skill beyond the calendar.

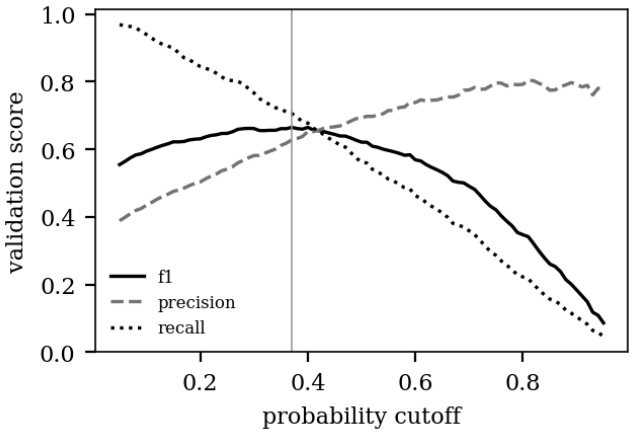


Figure 4. Phase 5 precision, recall and F1 against the probability cutoff; vertical line: tuned cutoff.

7. Bonus — asymmetric cost

Under-prediction was taken to cost three times as much as over-prediction of the same size, keeping the squared error: $\ell(r) = \frac{1}{2}c(r)r^2$ with $r = \hat{y} - y$ and $c = 3$ if $r < 0$, else 1. Its derivative $c(r)r$ is continuous, so the gradient is $X^T(c \odot r)/n = X^Tr/n + 2X^T\min(r, 0)/n$, and the gradient's Lipschitz constant is $3\lambda_{max}$; the minimiser is the 0.75 expectile rather than the mean. (A cost linear in the number of bikes would give the 75th percentile, but it has no gradient at zero, so fixed-step gradient descent could not converge on it.) On Phase 1's features the Phase 1 learning rate diverged within 12 updates, while the rate divided by three (0.318) converged in 388 updates. Predictions rose by 42.6 bikes per hour on average; R^2 fell from 0.682 to 0.606 while the asymmetric cost fell from 11,145 to 9,343; bikes short per hour fell from 37.2 to 21.2 and idle bikes rose from 39.6 to 66.1; the top-decile shortfall fell from 176.6 to 112.1. Much of the lift went to hours already over-predicted, a consequence of Phase 1's underfitting. On Phase 2's features the same loss shifted predictions by only 18.9 bikes, cost 0.016 R^2 (from 0.923) and reduced the top-decile shortfall from 58.9 to 34.1 bikes.

8. Limitations

(i) **Seasonal label leakage:** the Phase 5 label removes slot and year levels but not the season, so a large part of the classifier's ROC-AUC is reproducible from the calendar alone. (ii) **Chronological shortfall:** on later, busier days the models under-predict their own busiest hours (top-predicted-decile mean error -36.6 bikes for the target design, against -12.4 on the seeded split); growth beyond the training period is not extrapolated. (iii) **One validation set:** λ, the Phase 3 target and the Phase 5 cutoff were tuned and scored on the same seeded validation set, so those scores are slightly optimistic. (iv) **Grid revision:** the Phase 4 L1 paths were extended after the first official run because their best values lay at the grid edge; the revision is recorded in the notebook and changed only the Elastic Net choice. (v) The Phase 1 and bonus expectations were reconstructed after their runs and are labelled as such.

9. Final submission

The submitted predictions come from Phase 4's Ridge model ($\lambda = 0.00045$) on the 564-feature design. With every choice frozen on the seeded split, the model was refitted on all 10,886 labelled hours with the same procedure (encoder, scaler and centring means refitted on the combined rows; the same 92 exact combinations dropped). The final fit converged in 2,053 updates (training R^2 0.9317, RMSE 47.34). The 574 test hours were predicted in the sample file's order; 35 negative predictions were set to zero (mean 191.8, maximum 870.9 bikes).